BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 261 · Assembly language and addressing modes

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which primary memory is normally volatile?

Q2. What is the MAR used for?

Q3. What does an array index indicate?

Q4. For int[] a={5,8,3,7}, give a[2], then execute a[1]=a[1]+a[2]. State the new array and total.

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I Do · Read new knowledge

Key vocabulary

immediate — Operand value encoded as a literal.

direct — Memory address of data specified in instruction.

indirect — Specified memory holds a further effective address.

indexed — Base address modified by index or displacement.

effective address — Final memory location used for operand access.

New learning

Literal versus location — Immediate addressing embeds a literal value; direct addressing points to the memory address holding a value.

Follow a pointer for indirect — M[21] contains 20, so an indirect read through address 21 accesses M[20]=7.

Calculate indexed address — Base 20 plus index X=2 gives effective address 22 and value 3. These operand formats are generic examples rather than LMC syntax.

Worked example

M[30]=4, M[31]=30, M[32]=9, X=2. Evaluate #6, [30], [[31]], [30+X].

1. Predict — Immediate is literal 6; direct gets 4 from memory address 30.

2. Explain — Indirect follows pointer 30 stored at 31 to value 4; indexed adds offset 2 to base 30 to read 9.

We Do · Practise with guidance

M[40]=12, M[41]=42, M[42]=5, X=2. Evaluate #8, [40], [[41]], [40+X].

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#8=8; [40]=12
[[41]]=M[M[41]]=M[42]=5
[40+X]=M[42]=5
```

Predict

Immediate returns 8; direct reads M[40]=12.

Explain

M[41] points to address 42, so indirect reads 5. Indexed base 40+2 reads M[42]=5.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define immediate, direct, indirect and indexed addressing.

Question 2

Given M[10]=3, M[11]=10, M[12]=9 and X=2, find #4, [10], [[11]] and [10+X], explaining both address calculations.

Question 3

An array of 16-bit values begins at byte address 100. Find byte address of array[3] and explain why base+3 would be wrong for byte-addressed memory.

Exit Ticket · Answer before checking

Indirect addressing returns the pointer value itself rather than following the pointer.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. RAM.

Q2. Holding the address of memory being accessed.

Q3. The position of an element within an array.

Q4. a[2]=3; a[1]=11; array [5,11,3,7], sum 26.

You Do pages · Independent questions

Q1. [4] Immediate literal value in instruction [1]; direct location holds data [1]; indirect location holds another address [1]; indexed effective address is base+index [1].

Q2. [6] Immediate=4 [1]; direct M[10]=3 [1]; indirect first reads M[11]=10 [1], then M[10]=3 [1]; indexed address 10+2=12 [1], contents 9 [1].

Q3. [6] Array index 3 is fourth element [1]; each 16-bit element occupies 2 bytes [1]; displacement 3×2=6 bytes [2]; effective address 106 [1]; base+3 ignores element size [1].

Exit Ticket · Model answer

False. Indirect addressing uses the pointer to find the actual operand location.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding